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# OmegAMP: Targeted AMP Discovery via Biologically Informed Generation

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By 2050, **Antimicrobial Resistance** is expected to account for 2 million deaths each year

[Naddaf et al., 2024]

# Solution - Antimicrobial Peptides (AMPs)

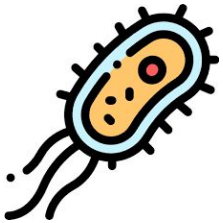
**Antimicrobial Peptide**

**(AMP)**

GLLKRIRFLL

kills

**Bacteria**

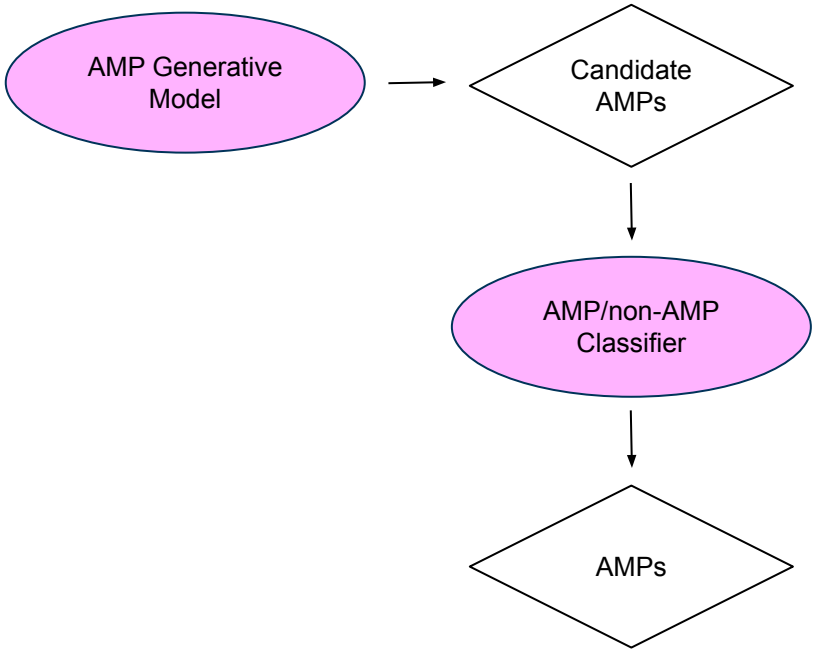


# AMP Discovery - Previous Approaches

- **Limited Controllability**
- **Ineffective Representation Spaces**
- **Unreliable Classification**

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- **Low Experimental Hit Rates**
- **Costly Discovery**




[Cao et al., 2023]



# OmegAMP Generation

# OmegAMP Generation

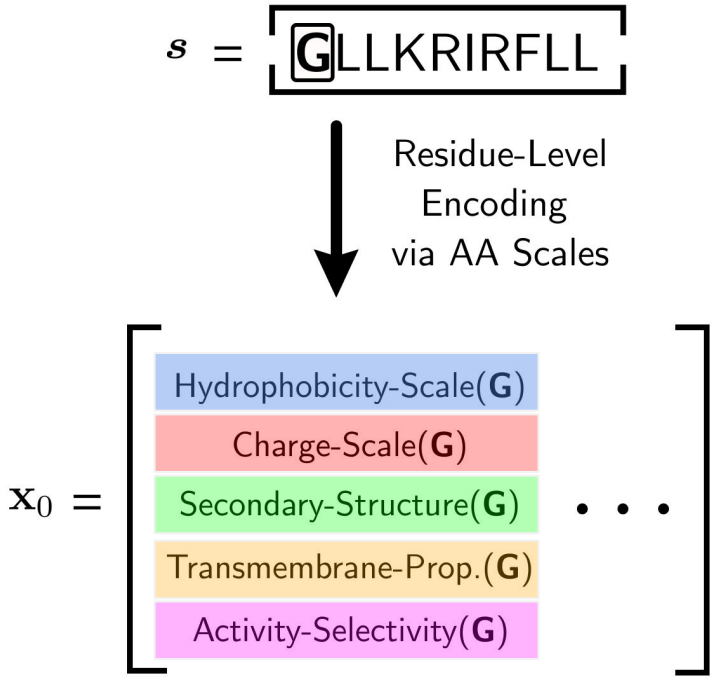
## Overview

- **Denoising Diffusion Model**
- **Conditional generation via input of conditioning representations**

## Core Contributions

- **Biologically-inspired embedding scheme**
- **Versatile conditioning mechanism**

# Biologically-inspired embedding via known Amino-acid scales




| AA | WW* | pI | Levitt | TM | AASI |
|---|---|---|---|---|---|
| A | -0.03 | 6.01 | 1.290 | 11.200 | 1.89 |
| R | -0.74 | 10.76 | 0.960 | 0.500 | 1.91 |
| N | -0.28 | 5.41 | 0.900 | 2.900 | 2.33 |
| D | -1.23 | 2.85 | 1.040 | 2.900 | 3.13 |
| C | 0.71 | 5.05 | 1.110 | 4.100 | 1.73 |
| Q | -0.51 | 5.65 | 1.270 | 1.600 | 3.05 |
| E | -2.02 | 3.15 | 1.440 | 1.800 | 3.14 |
| G | 0.37 | 6.06 | 0.560 | 11.800 | 2.67 |
| H | -0.89 | 6.00 | 1.220 | 2.000 | 3.00 |
| I | 0.81 | 6.05 | 0.970 | 8.600 | 1.97 |
| L | 1.06 | 6.01 | 1.300 | 11.700 | 1.74 |
| K | -0.99 | 9.60 | 1.230 | 0.500 | 2.28 |
| M | 0.61 | 5.74 | 1.470 | 1.900 | 2.50 |
| F | 1.63 | 5.49 | 1.070 | 5.100 | 1.53 |
| P | -0.38 | 6.30 | 0.520 | 2.700 | 0.22 |
| S | 0.17 | 5.68 | 0.820 | 8.000 | 2.14 |
| T | 0.07 | 5.60 | 0.820 | 4.900 | 2.18 |
| W | 2.35 | 5.89 | 0.990 | 2.200 | 2.00 |
| Y | 1.44 | 5.64 | 0.720 | 2.600 | 2.01 |
| V | 0.27 | 6.00 | 0.910 | 12.900 | 2.37 |

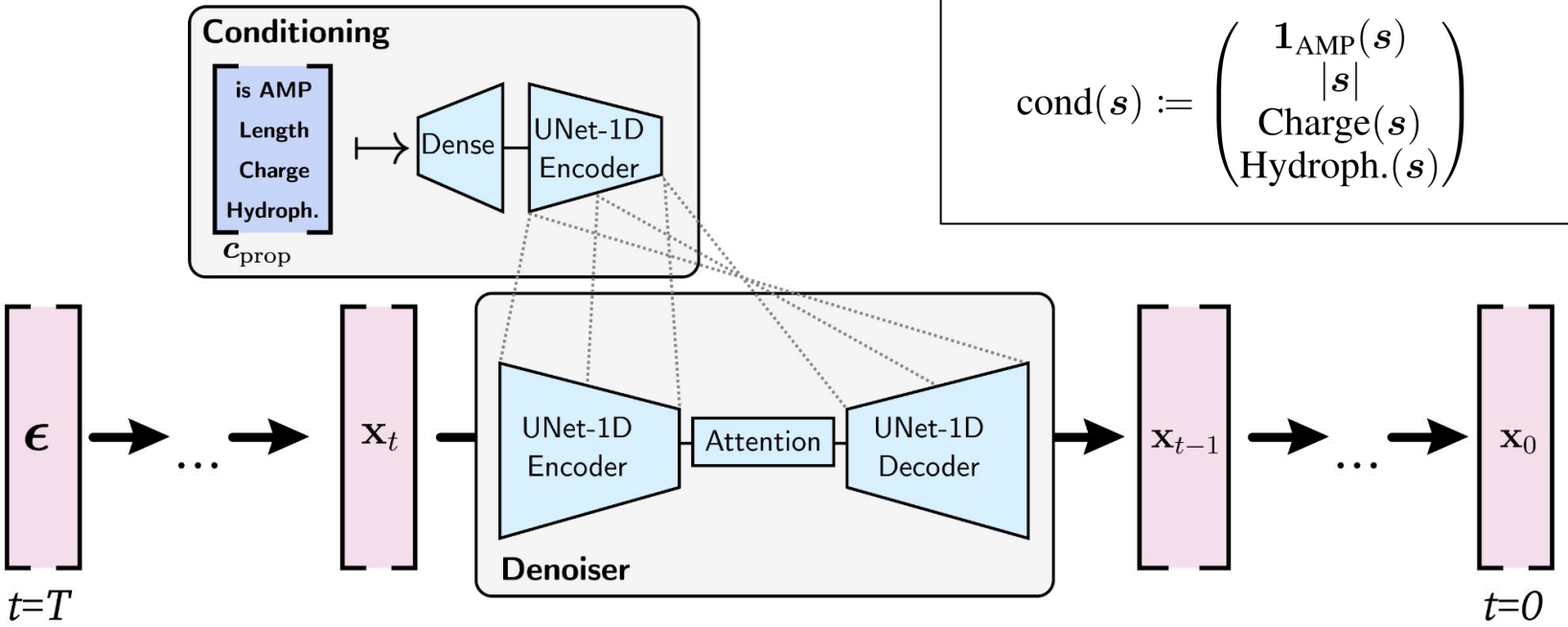
[Wimley et al., 1996], [Levitt et al., 1978], [Zhao et al., 2006],[Juretic et al., 2009]

# OmegAMP Generation
# -
# Conditioning Scheme

# Conditional Generation

**Conditional properties are precisely defined for every peptide**

$$\mathrm{cond}(s) := \begin{pmatrix} 1_{\mathrm{AMP}}(s) \\ |s| \\ \mathrm{Charge}(s) \\ \mathrm{Hydroph.}(s) \end{pmatrix}$$



# How to construct conditioning vectors with only partial information about target AMPs?

**(1) Property conditioning**

$$\boldsymbol{c}_{\text{prop}} = \begin{bmatrix} 1 \\ \ell \sim \mathcal{U}(\ell_{\min}, \ell_{\max}) \\ q \sim \mathcal{U}(q_{\min}, q_{\max}) \\ h \sim \mathcal{U}(h_{\min}, h_{\max}) \end{bmatrix}$$

**(2) Subset conditioning**

$$\boldsymbol{s} \sim \mathcal{S}_{E.coli}$$

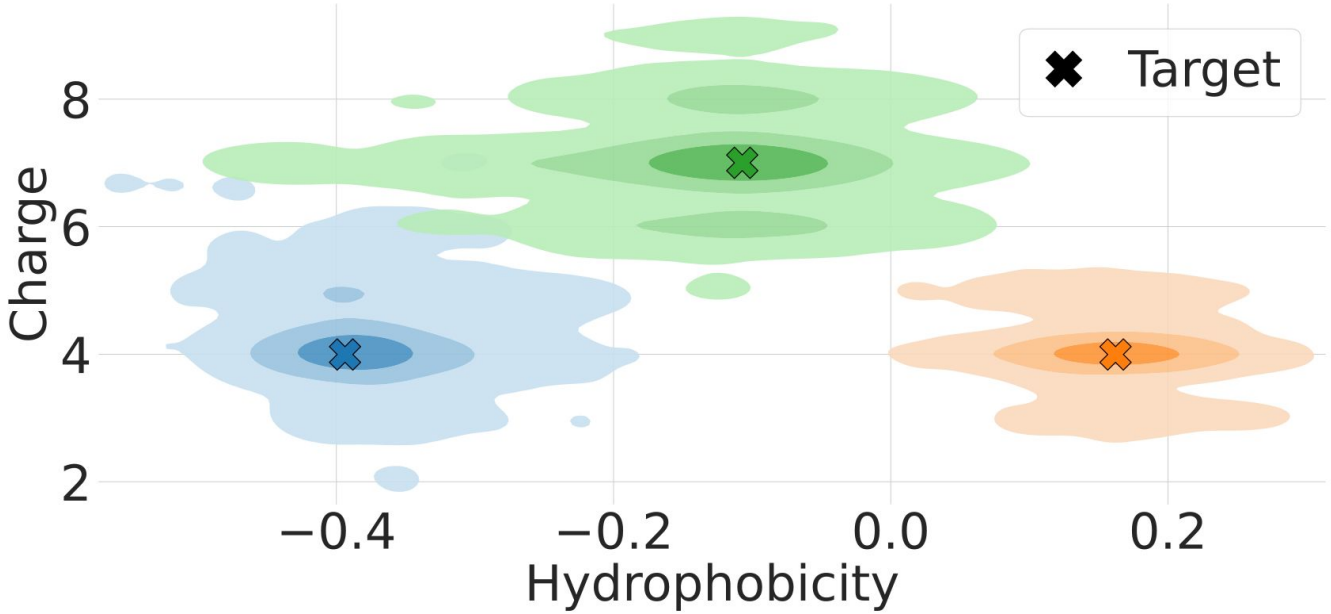
$$\boldsymbol{c}_{\text{prop}} = \text{cond}(\boldsymbol{s})$$

# Generation Experiments

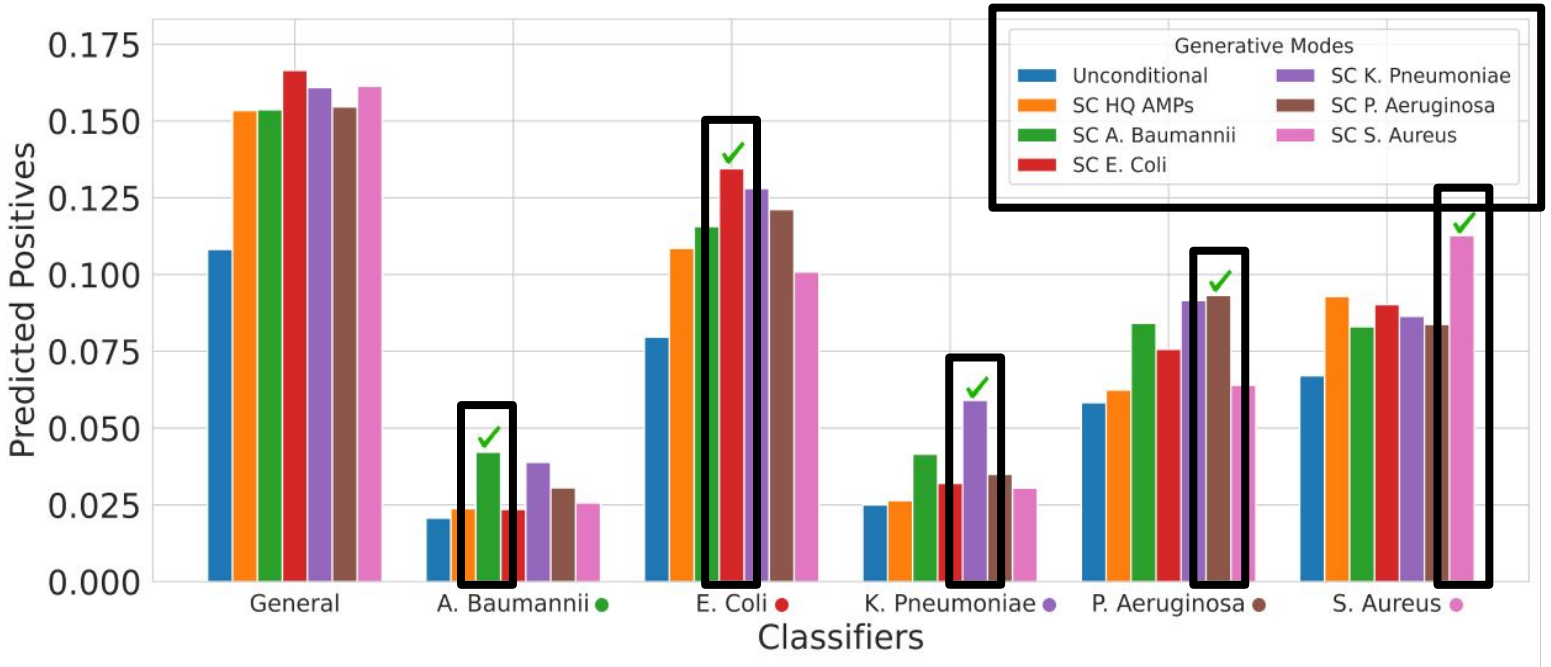
# OmegAMP generator achieves state-of-the-art performance

| Gen. Model | HydrAMP-MIC | OmegAMP Class. | Fitness Score | Diversity | Uniqueness |
|---|---|---|---|---|---|
| HQ AMPs (*Data*) | 81.5 | 42.4* | 0.16 | 0.62 | 100 |
| HQ non-AMPs (*Data*) | 20.5 | 6.3* | 0.13 | 0.63 | 100 |
| AMPGAN | 31.6 | 0.3 | 0.10 | 0.57 | **100** |
| Diff-AMP | 27.8 | 0.0 | 0.08 | 0.63 | **100** |
| HydrAMP | 44.1 | 0.0 | 0.09 | **0.70** | **100** |
| AMP-Diffusion | 42.8 | 2.2 | 0.11 | 0.64 | 91 |
| OmegAMP | 33.8 | 10.5 | 0.13 | 0.64 | 94 |
| OmegAMP-PC | **70.2** | 14.8 | **0.16** | 0.60 | 98 |
| OmegAMP-SC | 64.1 | **16.4** | 0.15 | 0.61 | 95 |

# Property conditioning closely follows physicochemical targets



# Subset conditioning enables bacteria-focused AMP generation

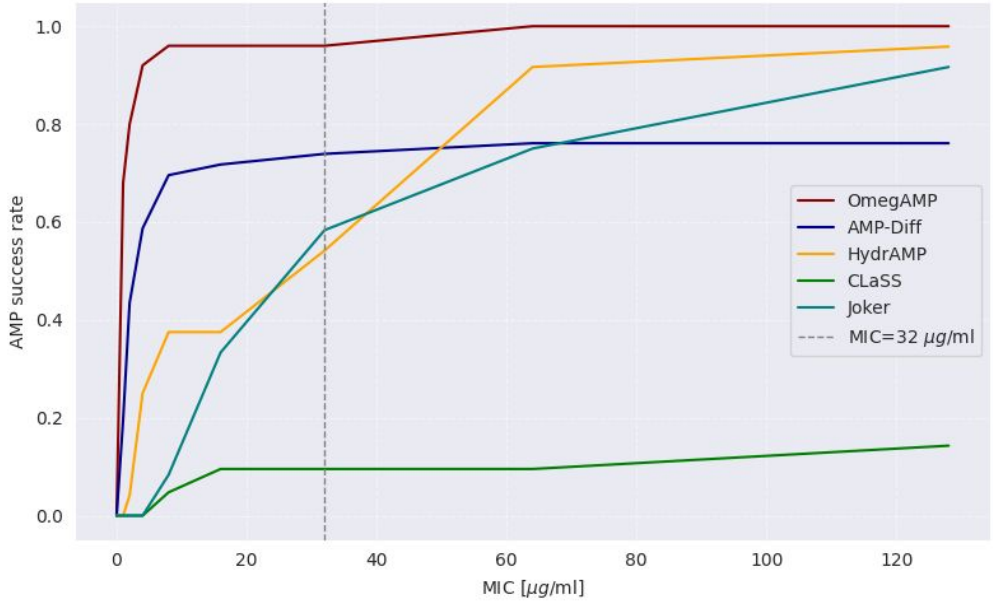


# *In vitro* experiments support *In silico* analysis

**Results**

**23 of 25 peptides active against multi-drug resistant bacteria**

**1 peptide active against all 17 considered strains**



# Conclusion

- **Controllable Generation -** first-in-class generative capabilities, allowing species focused and targeted physicochemical generation

- **Effective Biologically-Informed Representation -** as displayed by the superior modelling capabilities of our generative model

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# Thank you!











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